HK IMO Training 2021-2022
Problem Set 23
For 26 March 2022

Problem 89. There are n positive integers such that the sum of any two of them (the two integers should be distinct) is a power of 2. Find the largest possible value of n.

Problem 90. Let $P(x), Q(x)$ and $R(x)$ be nonzero polynomials with real coefficients such that

$$P(x) + Q(x) + R(x) = P(Q(x)) + Q(R(x)) + R(P(x)) = 0.$$

Prove that the degrees of all these polynomials are even.

Problem 91. Let $ABCD$ be a parallelogram where $\angle A$ is an acute angle. Let E be a point inside the parallelogram such that $AB = BE = CE$. The extension of the line CE meets the segment AD at F. Reflect the point F in the line DE to obtain the point F'. The line EF' meets the segment AD at P. Let Q be an intersection point of the circumcircles of $\triangle ADE$ and $\triangle CFF'$ such that A and Q lie on the same side of the line EF'. Prove that $\angle CQE = \angle DQP$.

Problem 92. Let n be an integer greater than 100. Prove that there are finitely many nonnegative integers k having the following property.

Property. Initially, we write down the integers $k+1, k+2, \ldots, k+n$ on the blackboard. In each step, we choose any two numbers a and b on the blackboard and replace them by one of the following numbers.

$$a+b, a-b, b-a, ab, \frac{a}{b}, \frac{b}{a}$$

(We can only do division if the denominator is nonzero.) After $n-1$ steps, it is possible to obtain the single number $n!$.